### 4.7.3 Graph of the inverse secant function

The inverse secant function, $y = \sec^{-1} x$ is a function whose domain is $\mathbb{R} \setminus (-1,1)$ and the range is $[0,\pi] \setminus \left\{\frac{\pi}{2}\right\}$. That is, $\sec^{-1} : \mathbb{R} \setminus (-1,1) \to [0,\pi] \setminus \left\{\frac{\pi}{2}\right\}$.

Fig. 4.25 and Fig. 4.26 are the graphs of the secant function in the principal domain and the inverse secant function in the corresponding domain, respectively.

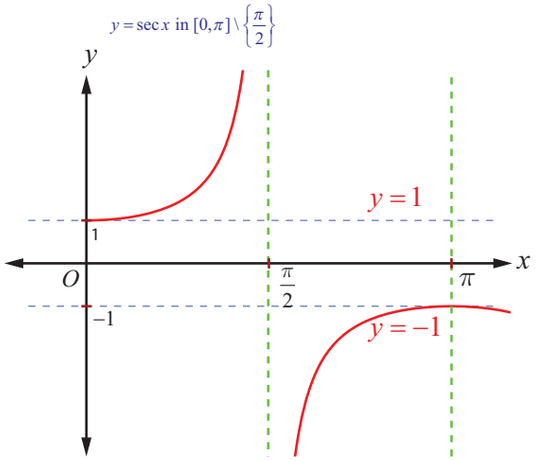



Fig. 4.25

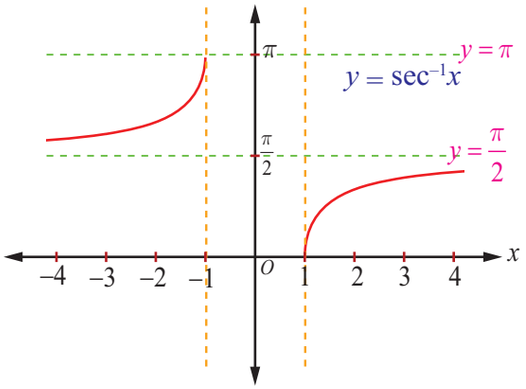



Fig. 4.26

**Remark**

A nice way to draw the graph of $y = \sec x$ or cosec $x$ :

(i) Draw the graph of $y = \cos x$ or $\sin x$

(ii) Draw the vertical asymptotes at the $x$-intercepts and take reciprocals of $y$ values.

## 4.8 The Cotangent Function and the Inverse Cotangent Function

The cotangent function is given by $\cot x = \frac{1}{\tan x}$. It is defined for all real values of $x$, except when $\tan x = 0$ or $x = n\pi, n \in \mathbb{Z}$. Thus, the domain of cotangent function is $\mathbb{R} \setminus \{n\pi : n \in \mathbb{Z}\}$ and its range is $(-\infty, \infty)$. Like $\tan x$, the cotangent function is an odd function and periodic with period $\pi$.

### 4.8.1 The graph of the cotangent function

The cotangent function is continuous on the set $(0,2\pi) \setminus \{\pi\}$. Let us first draw the graph of cotangent function in $(0,2\pi) \setminus \{\pi\}$. In the first and third quadrants, the cotangent function takes only positive values and in the second and fourth quadrants, it takes only negative values. The cotangent function has no maximum value and no minimum value. The cotangent function falls from $\infty$ to 0 for $x \in \left(0, \frac{\pi}{2}\right]$ ; falls from 0 to $-\infty$ for $x \in \left[\frac{\pi}{2}, \pi\right)$ ; falls from $\infty$ to 0 for $x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right]$ and falls from 0 to $-\infty$ for $x \in \left[\frac{3\pi}{2}, 2\pi\right)$.



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